PROBLEM 21 | Problems T2 defects.pdf

Crystal Defects & Solid Solutions

Proof of Dislocation Energy in FCC: [100] vs [110] Directions

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> *7. For a metal with FCC structure, E_1 and E_2 are the energies necessary to generate dislocations in the directions $[100]$ and $[110]$, respectively. Demonstrate that $E_1/E_2 = 2$.* *(Solution: theory)*

Solution: theory

PHASE 1: HYPOTHESIS & PARAMETERS

Lattice Model:

- Structure: Face-Centred Cubic (FCC), with lattice parameter a.
- Relation to the atomic radius: $a\sqrt{2} = 4R \implies a = 2\sqrt{2}R$.

Definition of Energies:

- E_1: Elastic energy per unit length to generate a dislocation with Burgers vector along the $[100]$ direction.
- E_2: Elastic energy per unit length to generate a dislocation with Burgers vector along the $[110]$ direction.

Quantity	Symbol	SI unit
Lattice parameter	a	m
Atomic radius	R	m
Burgers vector magnitude	$b = \lvert\vec{b}\rvert$	m
Dislocation energy per unit length	E	J/m
Proportionality constant ($E = k\,b^2$)	k	$\mathrm{J/m^3} = \mathrm{Pa}$ (of the order of the shear modulus)

Hypotheses: both dislocations are in the same FCC crystal, so k (which depends only on the elastic constants of the material and on the dislocation character, taken to be the same) is the same for both and cancels in the ratio; atoms are hard spheres touching along the face diagonal $\langle 110 \rangle$ [Session 4 Slide 36].

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Citing the mechanics of dislocations [Session 4 Slides 29, 36]:

1. **Elastic energy of a dislocation** (the course states it as "E used to move a dislocation $\propto |b|^2$" [Session 4 Slide 36]; in the literature this is the basis of Frank's criterion, an extension name not used on the slides):

The elastic strain energy stored in the elastic distortion field surrounding a dislocation is directly proportional to the **square of the magnitude of its Burgers vector**:

$$E \propto |\vec{b}|^2$$

2. **Determination of the Minimum Translation Vectors:**

For a dislocation to be kinematically stable in the crystal lattice, its Burgers vector must connect two equivalent lattice atomic positions:

- In the $[100]$ direction, the minimum translation joining two FCC lattice nodes (two adjacent vertices) is the full edge:

$$\vec{b}_1 = a[1\,0\,0]$$

- In the $[110]$ direction, owing to the presence of the face-centred atom, the shortest lattice translation goes from a vertex to the face centre:

$$\vec{b}_2 = \frac{a}{2}[1\,1\,0]$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Geometry of the FCC cell

In FCC the atoms touch along the face diagonal, which holds $4R$ (a corner atom contributes R, the face-centre atom $2R$, the opposite corner atom R). By Pythagoras on a face of side a:

$$a^2 + a^2 = (4R)^2 \implies 2a^2 = 16R^2 \implies a^2 = 8R^2 \implies a = 2\sqrt{2}\,R$$

2. Burgers vector along $[100]$

The shortest lattice translation along $[100]$ joins two neighbouring corners, $\vec{b}_1 = a[1\,0\,0] = (a, 0, 0)$ (the point $(\frac{1}{2}, 0, 0)$ is an octahedral interstice, not a lattice site). Its squared magnitude:

$$|\vec{b}_1|^2 = a^2 + 0^2 + 0^2 = a^2 = (2\sqrt{2}R)^2 = 4 \cdot 2 \cdot R^2 = 8R^2 \implies b_1 = 2\sqrt{2}\,R$$

This is the value on the slide, obtained there as $b^2 + b^2 = (4R)^2 \Rightarrow b = 2\sqrt{2}\,R$ [Session 4 Slide 36].

3. Burgers vector along $[110]$

Along $[110]$ the face-centre atom sits halfway along the diagonal, so the shortest translation is $\vec{b}_2 = \frac{a}{2}[1\,1\,0] = (\frac{a}{2}, \frac{a}{2}, 0)$:

$$|\vec{b}_2|^2 = \left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2 + 0^2 = \frac{a^2}{4} + \frac{a^2}{4} = \frac{2a^2}{4} = \frac{a^2}{2} = \frac{8R^2}{2} = 4R^2 \implies b_2 = 2R$$

This is one atomic diameter, the value on the slide ($b = 2R$) [Session 4 Slide 36].

4. Energies and their ratio

With $E = k\,b^2$ and the same k for both dislocations:

$$E_1 = k\,|\vec{b}_1|^2 = k\,a^2 = 8kR^2, \qquad E_2 = k\,|\vec{b}_2|^2 = k\,\frac{a^2}{2} = 4kR^2$$

$$\frac{E_1}{E_2} = \frac{k\,a^2}{k\,a^2/2} = a^2 \cdot \frac{2}{a^2} = 2 \qquad \text{and, in terms of } R: \qquad \frac{E_1}{E_2} = \frac{8kR^2}{4kR^2} = 2$$

$$\boxed{\frac{E_1}{E_2} = 2} \quad \blacksquare$$

The two routes (through a and through R) agree, and the result is the one requested by the statement.

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

1. **Dimensional check:** $[k\,b^2] = \mathrm{Pa} \cdot \mathrm{m}^2 = \frac{\mathrm{N}}{\mathrm{m}^2} \cdot \mathrm{m}^2 = \mathrm{N} = \frac{\mathrm{J}}{\mathrm{m}}$, an energy per unit length of dislocation line, as expected. The ratio $E_1/E_2 = \frac{\mathrm{J/m}}{\mathrm{J/m}}$ is dimensionless, and so is $b_1/b_2 = \frac{2\sqrt{2}R}{2R} = \sqrt{2}$; squaring it gives back $(\sqrt{2})^2 = 2$.

2. **Independence of the material:** a (or R) and k cancel, so the ratio 2 holds for every FCC metal (Al, Cu, Ni, ...), whatever its lattice parameter or shear modulus.
3. **Physical meaning:** the $[110]$ dislocation stores half the energy of the $[100]$ one because its Burgers vector is the shortest lattice translation, along the close-packed direction. This is why slip in FCC takes place along $\langle 110 \rangle$: dislocation movement is parallel to the direction of maximum packing [Session 4 Slide 36].
4. **Extension (not in the course material):** a $[100]$ dislocation can be written as the sum of two close-packed ones, $a[1\,0\,0] = \frac{a}{2}[1\,1\,0] + \frac{a}{2}[1\,\bar{1}\,0]$ (check: $\frac{a}{2}(1+1,\ 1-1,\ 0+0) = (a,0,0)$). The energies are $|a[100]|^2 = a^2$ before and $\frac{a^2}{2} + \frac{a^2}{2} = a^2$ after, so with $E \propto b^2$ alone this split is energy-neutral: the argument of the problem is the comparison of the two shortest translations, not a decomposition.